

DBS Lab 7 Tasks

1)

```

11 -- 1
12
13 SELECT E.FIRST_NAME, E.LAST_NAME, E.MANAGER_ID, M.FIRST_NAME AS MANAGER_FIRST_NAME, M.LAST_NAME AS MANAGER_LAST_NAME
14 FROM EMPLOYEES E
15 LEFT JOIN EMPLOYEES M
16 ON M.EMPLOYEE_ID = E.MANAGER_ID;
17

```

Query Result x

All Rows Fetched: 107 in 0.006 seconds

	FIRST_NAME	LAST_NAME	MANAGER_ID	MANAGER_FIRST_NAME	MANAGER_LAST_NAME
90	Mattea	Marvins	147	Alberto	Errazuriz
91	Danielle	Greene	147	Alberto	Errazuriz
92	Clara	Vishney	147	Alberto	Errazuriz
93	Sundita	Kumar	148	Gerald	Cambrault
94	Elizabeth	Bates	148	Gerald	Cambrault
95	William	Smith	148	Gerald	Cambrault
96	Taylor	Fox	148	Gerald	Cambrault
97	Harrison	Bloom	148	Gerald	Cambrault
98	Lisa	Ozer	148	Gerald	Cambrault
99	Charles	Johnson	149	Eleni	Zlotkey
100	Kimberely	Grant	149	Eleni	Zlotkey
101	Jack	Livingston	149	Eleni	Zlotkey
102	Jonathon	Taylor	149	Eleni	Zlotkey
103	Alyssa	Hutton	149	Eleni	Zlotkey
104	Ellen	Abel	149	Eleni	Zlotkey
105	Pat	Fay	201	Michael	Hartstein
106	William	Gietz	205	Shelley	Higgins
107	Steven	King	(null)	(null)	(null)

2)

```

7 -- 2
8 SELECT E.EMPLOYEE_ID, E.FIRST_NAME, E.SALARY, E.MANAGER_ID, M.FIRST_NAME AS MANAGER_FIRST_NAME, M.SALARY AS MANAGER_SALARY
9 FROM EMPLOYEES E
10 JOIN EMPLOYEES M
11 ON M.EMPLOYEE_ID = E.MANAGER_ID
12 WHERE E.SALARY > M.SALARY;

```

Query Result x

All Rows Fetched: 2 in 0.001 seconds

	EMPLOYEE_ID	FIRST_NAME	SALARY	MANAGER_ID	MANAGER_FIRST_NAME	MANAGER_SALARY
1	168	Lisa	11500	148	Gerald	11000
2	174	Ellen	11000	149	Eleni	10500

3)

```

14  -- 3
15  SELECT D.DEPARTMENT_ID, D.DEPARTMENT_NAME FROM DEPARTMENTS D
16  LEFT JOIN EMPLOYEES E
17  ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
18  WHERE E.EMPLOYEE_ID IS NULL;

```

Query Result x

SQL | All Rows Fetched: 16 in 0.007 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME
1	120	Treasury
2	130	Corporate Tax
3	140	Control And Credit
4	150	Shareholder Services
5	160	Benefits
6	170	Manufacturing
7	180	Construction
8	190	Contracting
9	200	Operations
10	210	IT Support
11	220	NOC
12	230	IT Helpdesk
13	240	Government Sales
14	250	Retail Sales
15	260	Recruiting
16	270	Payroll

4)

```

20  -- 4
21  SELECT E.EMPLOYEE_ID, E.HIRE_DATE, M.EMPLOYEE_ID AS MANAGER_ID, M.HIRE_DATE AS MANAAGER_HIRE FROM EMPLOYEES E
22  JOIN EMPLOYEES M
23  ON E.MANAGER_ID = M.EMPLOYEE_ID
24  WHERE E.HIRE_DATE > M.HIRE_DATE;

```

Query Result x

SQL | All Rows Fetched: 68 in 0.005 seconds

	EMPLOYEE_ID	HIRE_DATE	MANAGER_ID	MANAAGER_HIRE
1	20117	FEB-04	10017	JUN-03
2	14929	JAN-08	10017	JUN-03
3	14815	OCT-07	10017	JUN-03
4	14710	MAR-05	10017	JUN-03
5	14605	JAN-05	10017	JUN-03
6	14501	OCT-04	10017	JUN-03
7	12416	NOV-07	10017	JUN-03
8	12310	OCT-05	10017	JUN-03
9	12110	APR-05	10017	JUN-03
10	12018	JUL-04	10017	JUN-03
11	10121	SEP-05	10017	JUN-03
12	10303	JAN-06	10213	JAN-01
13	10707	FEB-07	10303	JAN-06
14	10605	FEB-06	10303	JAN-06
15	10421	MAY-07	10303	JAN-06
16	11307	DEC-07	10817	AUG-02
17	11207	MAR-06	10817	AUG-02
18	11130	SEP-05	10817	AUG-02

5)

```

26 -- 5
27 SELECT E1.EMPLOYEE_ID, E1.FIRST_NAME, E1.LAST_NAME, E2.EMPLOYEE_ID, E2.FIRST_NAME, E2.LAST_NAME
28 FROM EMPLOYEES E1 JOIN EMPLOYEES E2
29 ON E1.DEPARTMENT_ID = E2.DEPARTMENT_ID AND E1.EMPLOYEE_ID != E2.EMPLOYEE_ID;

```

Script Output x Query Result x

SQL | Fetched 1,200 rows in 0.031 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMPLOYEE_ID_1	FIRST_NAME_1	LAST_NAME_1
1	102	Lex	De Haan	100	Steven	King
2	101	Neena	Kochhar	100	Steven	King
3	102	Lex	De Haan	101	Neena	Kochhar
4	100	Steven	King	101	Neena	Kochhar
5	101	Neena	Kochhar	102	Lex	De Haan
6	100	Steven	King	102	Lex	De Haan
7	107	Diana	Lorentz	103	Alexander	Hunold
8	106	Valli	Pataballa	103	Alexander	Hunold
9	105	David	Austin	103	Alexander	Hunold
10	104	Bruce	Ernst	103	Alexander	Hunold
11	107	Diana	Lorentz	104	Bruce	Ernst
12	106	Valli	Pataballa	104	Bruce	Ernst
13	105	David	Austin	104	Bruce	Ernst
14	103	Alexander	Hunold	104	Bruce	Ernst
15	107	Diana	Lorentz	105	David	Austin
16	106	Valli	Pataballa	105	David	Austin
17	104	Bruce	Ernst	105	David	Austin
18	103	Alexander	Hunold	105	David	Austin

6)

```

31  -- 6
32  SELECT E.FIRST_NAME, E.LAST_NAME, D.DEPARTMENT_NAME, L.CITY
33  FROM EMPLOYEES E
34  JOIN DEPARTMENTS D
35       ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
36  JOIN LOCATIONS L
37       ON D.LOCATION_ID = L.LOCATION_ID
38  ;
39

```

Script Output x Query Result x				
SQL All Rows Fetched: 106 in 0.005 seconds				
	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME	CITY
1	Ellen	Abel	Sales	Oxford
2	Sundar	Ande	Sales	Oxford
3	Mozhe	Atkinson	Shipping	South San Francisco
4	David	Austin	IT	Southlake
5	Hermann	Baer	Public Relations	Munich
6	Shelli	Baida	Purchasing	Seattle
7	Amit	Banda	Sales	Oxford
8	Elizabeth	Bates	Sales	Oxford
9	Sarah	Bell	Shipping	South San Francisco
10	David	Bernstein	Sales	Oxford
11	Laura	Bissot	Shipping	South San Francisco
12	Harrison	Bloom	Sales	Oxford
13	Alexis	Bull	Shipping	South San Francisco
14	Anthony	Cabrio	Shipping	South San Francisco
15	Gerald	Cambrault	Sales	Oxford
16	Nanette	Cambrault	Sales	Oxford
17	John	Chen	Finance	Seattle
18	Kelly	Chung	Shipping	South San Francisco

7)

```

40  -- 7
41  SELECT E1.EMPLOYEE_ID, E1.FIRST_NAME, E1.LAST_NAME, E1.DEPARTMENT_ID
42  FROM EMPLOYEES E1
43  JOIN EMPLOYEES E2
44      ON E1.DEPARTMENT_ID = E2.DEPARTMENT_ID
45      WHERE E2.FIRST_NAME = 'Steven'
46          AND E2.LAST_NAME = 'King'
47          AND E1.EMPLOYEE_ID != E2.EMPLOYEE_ID
48  ;

```

Script Output x Query Result x

SQL | All Rows Fetched: 2 in 0.003 seconds

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	DEPARTMENT_ID
1	101 Neena	Kochhar	90
2	102 Lex	De Haan	90

8)

```

50  -- 8
51  SELECT E.EMPLOYEE_ID, E.FIRST_NAME, E.LAST_NAME, E.SALARY, E.DEPARTMENT_ID
52  FROM EMPLOYEES E
53  JOIN (
54      SELECT DEPARTMENT_ID, AVG(SALARY) AS DEPT_AVG
55      FROM EMPLOYEES
56      GROUP BY DEPARTMENT_ID
57  ) T
58      ON E.DEPARTMENT_ID = T.DEPARTMENT_ID AND E.SALARY > T.DEPT_AVG
59  ;

```

Script Output x Query Result x

SQL | All Rows Fetched: 38 in 0.005 seconds

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
1	100 Steven	King	24000	90
2	103 Alexander	Hunold	9000	60
3	104 Bruce	Ernst	6000	60
4	108 Nancy	Greenberg	12008	100
5	109 Daniel	Faviet	9000	100
6	114 Den	Raphaely	11000	30
7	120 Matthew	Weiss	8000	50
8	121 Adam	Fripp	8200	50
9	122 Payam	Kaufling	7900	50
10	123 Shanta	Vollman	6500	50
11	124 Kevin	Mourgos	5800	50
12	137 Renske	Ladwig	3600	50
13	141 Tenna	Rajs	3500	50
14	145 John	Russell	14000	80
15	146 Karen	Partners	13500	80
16	147 Alberto	Errazuriz	12000	80
17	148 Gerald	Cambrault	11000	80
18	149 Eleni	Zlotkey	10500	80

9)

```

61 -- 9
62 SELECT E.EMPLOYEE_ID, E.FIRST_NAME, E.LAST_NAME, COALESCE(D.DEPARTMENT_NAME, 'NONE')
63 FROM EMPLOYEES E
64 LEFT OUTER JOIN DEPARTMENTS D
65 ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
66 ;
67

```

Script Output x Query Result x

SQL | All Rows Fetched: 107 in 0.005 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	COALESCE(D.DEPARTMENT_NAME,'NONE')
90	150	Peter	Tucker	Sales
91	149	Eleni	Zlotkey	Sales
92	148	Gerald	Cambrault	Sales
93	147	Alberto	Errazuriz	Sales
94	146	Karen	Partners	Sales
95	145	John	Russell	Sales
96	102	Lex	De Haan	Executive
97	101	Neena	Kochhar	Executive
98	100	Steven	King	Executive
99	113	Luis	Popp	Finance
100	112	Jose M...	Urman	Finance
101	111	Ismael	Sciarra	Finance
102	110	John	Chen	Finance
103	109	Daniel	Faviet	Finance
104	108	Nancy	Greenberg	Finance
105	206	William	Gietz	Accounting
106	205	Shelley	Higgins	Accounting
107	178	Kimberely	Grant	NONE

10)

```

68 -- 10
69 SELECT E.EMPLOYEE_ID, E.LAST_NAME
70 FROM EMPLOYEES E
71 LEFT JOIN JOB_HISTORY J ON E.EMPLOYEE_ID = J.EMPLOYEE_ID
72 WHERE J.EMPLOYEE_ID IS NULL;

```

Query Result x

SQL | All Rows Fetched: 100 in 0.087 seconds

	EMPLOYEE_ID	LAST_NAME
1	174	Abel
2	166	Ande
3	130	Atkinson
4	105	Austin
5	204	Baer
6	116	Baida
7	167	Banda
8	172	Bates
9	192	Bell
10	151	Bernstein
11	129	Bissot
12	169	Bloom
13	185	Bull

11)

```
74  -- 11
75  SELECT JOB_ID FROM EMPLOYEES
76  UNION
77  SELECT JOB_ID FROM JOB_HISTORY;
```

Query Result x

SQL | All Rows Fetched: 19 in 0.012 seconds

	JOB_ID
1	AC_ACCOUNT
2	AC_MGR
3	AD_ASST
4	AD_PRES
5	AD_VP
6	FI_ACCOUNT
7	FI_MGR
8	HR_REP
9	IT_PROG
10	MK_MAN
11	MK_REP
12	PR_REP
13	PU_CLERK

12)

```
79  -- 12
80  SELECT JOB_ID FROM EMPLOYEES
81  MINUS
82  SELECT JOB_ID FROM JOB_HISTORY;
```

Query Result x

SQL | All Rows Fetched: 11 in 0 seconds

	JOB_ID
1	AD_PRES
2	AD_VP
3	FI_ACCOUNT
4	FI_MGR
5	HR_REP
6	MK_MAN
7	PR_REP
8	PU_CLERK
9	PU_MAN
10	SH_CLERK
11	ST_MAN

13)


```

84  -- 13
85  SELECT EMPLOYEE_ID FROM EMPLOYEES
86  INTERSECT
87  SELECT EMPLOYEE_ID FROM JOB_HISTORY;

```

Query Result x

SQL | All Rows Fetched: 7 in 0.004 seconds

	EMPLOYEE_ID
1	101
2	102
3	114
4	122
5	176
6	200
7	201

14)

```

89  -- 14
90  SELECT DEPARTMENT_ID FROM EMPLOYEES
91  INTERSECT
92  SELECT DEPARTMENT_ID FROM JOB_HISTORY;

```

Query Result x

SQL | All Rows Fetched: 6 in 0.007 seconds

	DEPARTMENT_ID
1	20
2	50
3	60
4	80
5	90
6	110

15)

```

94  -- 15
95  SELECT EMPLOYEE_ID FROM EMPLOYEES
96  UNION
97  SELECT EMPLOYEE_ID FROM JOB_HISTORY;

```

Query Result x

SQL | All Rows Fetched: 107 in 0.005 seconds

	EMPLOYEE_ID
1	100
2	101
3	102
4	103
5	104
6	105
7	106
8	107
9	108
10	109
11	110
12	111
13	112

16)

```

99  -- 16
100 SELECT EMPLOYEE_ID FROM JOB_HISTORY
101 MINUS
102 SELECT EMPLOYEE_ID FROM EMPLOYEES;

```

Query Result x

SQL | All Rows Fetched: 0 in 0.002 seconds

EMPLOYEE...

17)

```

104  -- 17
105  SELECT JOB_ID FROM JOB_HISTORY
106  UNION ALL
107  SELECT JOB_ID FROM EMPLOYEES;

```

Query Result x

SQL | All Rows Fetched: 117 in 0.014 seconds

	JOB_ID
1	AC_ACCOUNT
2	AC_ACCOUNT
3	AC_MGR
4	AD_ASST
5	IT_PROG
6	MK_REP
7	SA_MAN
8	SA_REP
9	ST_CLERK
10	ST_CLERK
11	AC_ACCOUNT
12	AC_MGR
13	AD_ASST

18)

```
109  -- 18
110  SELECT DEPARTMENT_ID
111  FROM DEPARTMENTS
112  WHERE DEPARTMENT_ID IN (
113      SELECT DEPARTMENT_ID FROM EMPLOYEES
114      INTERSECT
115      SELECT DEPARTMENT_ID FROM JOB_HISTORY
116  );
```

Query Result x	
All Rows Fetched: 6 in 0.005 seconds	
	DEPARTMENT_ID
1	20
2	50
3	60
4	80
5	90
6	110

19)

117	
118	-- 19
119	SELECT EMPLOYEE_ID FROM EMPLOYEES
120	MINUS
121	SELECT EMPLOYEE_ID FROM JOB_HISTORY;

Query Result x	
SQL All Rows Fetched: 100 in 0.003 seconds	
	EMPLOYEE_ID
1	100
2	103
3	104
4	105
5	106
6	107
7	108
8	109
9	110
10	111
11	112
12	113
13	115

20)

```

123 -- 20
124 SELECT EMPLOYEE_ID, JOB_ID FROM EMPLOYEES
125 MINUS
126 SELECT EMPLOYEE_ID, JOB_ID FROM JOB_HISTOR

```

Query Result x

SQL | All Rows Fetched: 105 in 0.013 seconds

	EMPLOYEE_ID	JOB_ID
1	100	AD_PRES
2	101	AD_VP
3	102	AD_VP
4	103	IT_PROG
5	104	IT_PROG
6	105	IT_PROG
7	106	IT_PROG
8	107	IT_PROG
9	108	FI_MGR
10	109	FI_ACCOUNT
11	110	FI_ACCOUNT
12	111	FI_ACCOUNT
13	112	FI_ACCOUNT

21)

```

128 -- 21
129 CREATE VIEW EMP_DEPT_SAL AS
130     SELECT E.FIRST_NAME || ' ' || E.LAST_NAME AS FULL_NAME, D.DEPARTMENT_NAME, E.SALARY
131     FROM EMPLOYEES E
132     JOIN DEPARTMENTS D
133     ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
134 ;

```

Script Output x Query Result x

Task completed in 0.039 seconds

```

FROM EMPLOYEES E
JOIN DEPARTMENTS D
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
Error report -
ORA-00904: "E"."LASTNAME": invalid identifier
00904. 00000 - "%s: invalid identifier"
*Cause:
*Action:

View EMP_DEPT_SAL created.

```

22)

```
136 -- 22
137 CREATE VIEW HIGH_EARNERS AS
138     SELECT * FROM EMPLOYEES
139     WHERE SALARY > 10000
140 ;
```

Script Output x Query Result x

Task completed in 0.02 seconds

Error report -
ORA-00904: "E"."LASTNAME": invalid identifier
00904. 00000 - "%s: invalid identifier"
*Cause:
*Action:

View EMP_DEPT_SAL created.

View HIGH_EARNERS created.

23)

```
142 -- 23
143 CREATE VIEW HIERARCHY_REPORT AS
144     SELECT E.FIRST_NAME || ' ' || E.LAST_NAME AS EMPLOYEE, M.FIRST_NAME || ' ' || M.LAST_NAME AS MANAGER
145     FROM EMPLOYEES E
146     JOIN EMPLOYEES M
147     ON E.MANAGER_ID = M.EMPLOYEE_ID
148 ;
```

Script Output x Query Result x

Task completed in 0.036 seconds

*Cause:
*Action:

View EMP_DEPT_SAL created.

View HIGH_EARNERS created.

View HIERARCHY_REPORT created.

24)

```
150  -- 24
151  CREATE VIEW DEPT_SAL_SUMM AS
152      SELECT DEPARTMENT_ID, AVG(SALARY) AS AVG_SAL
153      FROM EMPLOYEES
154      GROUP BY DEPARTMENT_ID
155  ;
```

Script Output x Query Result x
Task completed in 0.04 seconds

View EMP_DEPT_SAL created.

View HIGH_EARNERS created.

View HIERARCHY_REPORT created.

View DEPT_SAL_SUMM created.

25)


```
157 -- 25
158 CREATE VIEW EMP_DEPT_LOC_DEETS AS
159     SELECT E.LAST_NAME, D.DEPARTMENT_NAME, L.CITY
160     FROM EMPLOYEES E
161     JOIN DEPARTMENTS D
162         ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
163     JOIN LOCATIONS L
164         ON D.LOCATION_ID = L.LOCATION_ID
165 ;
```

Script Output x Query Result x
Task completed in 0.038 seconds

View HIGH_EARNERS created.

View HIERARCHY_REPORT created.

View DEPT_SAL_SUMM created.

View EMP_DEPT_LOC_DEETS created.

26)

```
167  -- 26
168  CREATE VIEW SECURE_SALARY_VIEW AS
169      SELECT * FROM EMPLOYEES
170      WHERE SALARY > 5000
171  WITH CHECK OPTION;
```

Script Output x Query Result x
Task completed in 0.031 seconds

View HIERARCHY_REPORT created.

View DEPT_SAL_SUMM created.

View EMP_DEPT_LOC_DEETS created.

View SECURE_SALARY_VIEW created.

27)

```
173  -- 27
174  CREATE VIEW JOB_HISTORY_READONLY AS
175      SELECT * FROM JOB_HISTORY
176  WITH READ ONLY;
```

Script Output x Query Result x
Task completed in 0.033 seconds

View DEPT_SAL_SUMM created.

View EMP_DEPT_LOC_DEETS created.

View SECURE_SALARY_VIEW created.

View JOB_HISTORY_READONLY created.

28)

```
178 -- 28
179 CREATE VIEW UNIFIED_JOBS AS
180     SELECT EMPLOYEE_ID, JOB_ID FROM EMPLOYEES
181     UNION
182     SELECT EMPLOYEE_ID, JOB_ID FROM JOB_HISTORY
183 ;
```

Script Output x Query Result x

Task completed in 0.026 seconds

View EMP_DEPT_LOC_DEETS created.

View SECURE_SALARY_VIEW created.

View JOB_HISTORY_READONLY created.

View UNIFIED_JOBS created.

29)

```
184
185 -- 29
186 UPDATE HIGH_EARNERS
187     SET SALARY = SALARY + 500
188     WHERE EMPLOYEE_ID = 101
189     -- success bcuz high_earners is a simple view based
190     -- on one table without aggregations.
191 ;
192
```

Script Output x Query Result x

Task completed in 0.024 seconds

View SECURE_SALARY_VIEW created.

View JOB_HISTORY_READONLY created.

View UNIFIED_JOBS created.

1 row updated.

30)

```
193 -- 30
194 CREATE VIEW ABOVE_AVG_DEPT AS
195     SELECT * FROM EMPLOYEES E
196     WHERE SALARY > (
197         SELECT AVG(SALARY) FROM EMPLOYEES
198         WHERE DEPARTMENT_ID = E.DEPARTMENT_ID
199     )
200 ;
```

Script Output x

Query Result x

Task completed in 0.032 seconds

View JOB_HISTORY_READONLY created.

View UNIFIED_JOBS created.

1 row updated.

View ABOVE_AVG_DEPT created.

31)

```

202  -- 31
203 CREATE VIEW MANAGER_DEPT_CHECK AS
204     SELECT E.LAST_NAME AS EMP,
205            E.DEPARTMENT_ID AS EMP_DEPT,
206            M.LAST_NAME AS MGR,
207            M.DEPARTMENT_ID AS MGR_DEPT
208     FROM EMPLOYEES E JOIN EMPLOYEES M
209          ON E.MANAGER_ID = M.EMPLOYEE_ID
210 ;
211
212 SELECT * FROM MANAGER_DEPT_CHECK
213 WHERE EMP_DEPT != MGR_DEPT;

```

Script Output x Query Result x				
SQL All Rows Fetched: 18 in 0 seconds				
	EMP	EMP_DEPT	MGR	MGR_DEPT
1	Hartstein	20	King	90
2	Zlotkey	80	King	90
3	Cambrault	80	King	90
4	Errazuriz	80	King	90
5	Partners	80	King	90
6	Russell	80	King	90
7	Mourgos	50	King	90
8	Vollman	50	King	90
9	Kaufling	50	King	90
10	Fripp	50	King	90
11	Weiss	50	King	90
12	Raphaely	30	King	90
13	Higgins	110	Neenan	90

32)

```

215 -- 32
216 CREATE VIEW EMP_CITY_VIEW AS
217     SELECT E.EMPLOYEE_ID, L.CITY FROM EMPLOYEES E
218     JOIN DEPARTMENTS D
219         ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
220     JOIN LOCATIONS L
221         ON D.LOCATION_ID = L.LOCATION_ID
222 ;
223
224 SELECT * FROM EMP_CITY_VIEW
225 WHERE CITY IN (
226     SELECT CITY FROM EMP_CITY_VIEW
227     GROUP BY CITY
228     HAVING COUNT(*) > 5
229 );

```

Script Output x Query Result x

    SQL | All Rows Fetched: 97 in 0.011 seconds

	EMPLOYEE_ID	CITY
1	200	Seattle
2	114	Seattle
3	115	Seattle
4	116	Seattle
5	117	Seattle
6	118	Seattle
7	119	Seattle
8	100	Seattle
9	101	Seattle
10	102	Seattle

33)

35)

```
254 -- 35
255 CREATE VIEW DEPT_COUNTS AS
256     SELECT DEPARTMENT_ID, COUNT(*) AS EMP_COUNT
257     FROM EMPLOYEES
258     GROUP BY DEPARTMENT_ID
259 ;
260
261 SELECT * FROM DEPT_COUNTS
262     WHERE EMP_COUNT > (
263         SELECT AVG(COUNT(*)) FROM EMPLOYEES
264         GROUP BY DEPARTMENT_ID
265     )
266 ;
```

Script Output x Query Result x

SQL | All Rows Fetched: 2 in 0 seconds

	DEPARTMENT_ID	EMP_COUNT
1	50	45
2	80	34

36)

```
268 -- 36
269 CREATE VIEW SAL_COMPARE AS
270     SELECT E.LAST_NAME AS EMP, E.SALARY AS E_SAL, M.SALARY AS M_SAL
271     FROM EMPLOYEES E
272     JOIN EMPLOYEES M
273         ON E.MANAGER_ID = M.EMPLOYEE_ID
274 ;
275
276 SELECT * FROM SAL_COMPARE
277     WHERE E_SAL > M_SAL;
```

Script Output x Query Result x

SQL | All Rows Fetched: 2 in 0.005 seconds

	EMP	E_SAL	M_SAL
1	Ozer	11500	11000
2	Abel	11000	10500

37)

```

279 -- 37
280 CREATE VIEW EMP_MGR_LOC AS
281     SELECT E.LAST_NAME, L.CITY AS EMP_CITY, LM.CITY AS MGR_CITY
282     FROM EMPLOYEES E
283     JOIN DEPARTMENTS D
284         ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
285     JOIN LOCATIONS L
286         ON D.LOCATION_ID = L.LOCATION_ID
287     JOIN EMPLOYEES M
288         ON E.MANAGER_ID = M.EMPLOYEE_ID
289     JOIN DEPARTMENTS DM
290         ON M.DEPARTMENT_ID = DM.DEPARTMENT_ID
291     JOIN LOCATIONS LM
292         ON DM.LOCATION_ID = LM.LOCATION_ID
293 ;
294
295 SELECT * FROM EMP_MGR_LOC WHERE EMP_CITY = MGR_CITY;

```

Script Output x Query Result x

SQL | All Rows Fetched: 91 in 0.014 seconds

	LAST_NAME	EMP_CITY	MGR_CITY
1	Raphaely	Seattle	Seattle
2	De Haan	Seattle	Seattle
3	Kochhar	Seattle	Seattle
4	Higgins	Seattle	Seattle
5	Whalen	Seattle	Seattle
6	Greenberg	Seattle	Seattle
7	Lorentz	Southlake	Southlake

38)

```

297 -- 38
298 CREATE VIEW ROLE_CHECK AS
299     SELECT E.EMPLOYEE_ID, E.JOB_ID AS CURRENT_JOB, J.JOB_ID AS PAST_JOB
300     FROM EMPLOYEES E
301     JOIN JOB_HISTORY J
302         ON E.EMPLOYEE_ID = J.EMPLOYEE_ID
303 ;
304
305 SELECT DISTINCT EMPLOYEE_ID FROM ROLE_CHECK
306 WHERE CURRENT_JOB = PAST_JOB;

```

Script Output x Query Result x

SQL | All Rows Fetched: 2 in 0.007 seconds

	EMPLOYEE_ID
1	200
2	176

39)

```

308 -- 39
309 CREATE VIEW HIRE_DATE_COMPARE AS
310     SELECT E.LAST_NAME, E.HIRE_DATE AS EMP_HIRE, M.HIRE_DATE AS MGR_HIRE
311     FROM EMPLOYEES E
312     JOIN EMPLOYEES M
313         ON E.MANAGER_ID = M.EMPLOYEE_ID
314 ;
315
316 SELECT * FROM HIRE_DATE_COMPARE
317 WHERE EMP_HIRE < MGR_HIRE;

```

Script Output x Query Result x

SQL | All Rows Fetched: 37 in 0.003 seconds

	LAST_NAME	EMP_HIRE	MGR_HIRE
1	Kaufling	01-MAY-03	17-JUN-03
2	Raphaely	07-DEC-02	17-JUN-03
3	De Haan	13-JAN-01	17-JUN-03
4	Higgins	07-JUN-02	21-SEP-05
5	Baer	07-JUN-02	21-SEP-05
6	Mavris	07-JUN-02	21-SEP-05
7	Whalen	17-SEP-03	21-SEP-05

40)

```

319 -- 40
320 CREATE VIEW EMP_DEPT_JOB_DEETS AS
321     SELECT E.EMPLOYEE_ID, E.SALARY, D.DEPARTMENT_NAME, J.JOB_TITLE
322     FROM EMPLOYEES E
323     JOIN DEPARTMENTS D
324         ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
325     JOIN JOBS J
326         ON E.JOB_ID = J.JOB_ID
327 ;
328
329 UPDATE EMP_DEPT_JOB_DEETS
330     SET SALARY = 9000
331     WHERE EMPLOYEE_ID = 101;
332 -- Operation succeeds
333 -- Comparing with a primary key ensures that the data exists within
334 -- that table, allowing updates, on a table with keys.

```

Script Output x Query Result x

Task completed in 0.038 seconds

View EMP_DEPT_JOB_DEETS created.

1 row updated.